

Experiment 2: Charging and discharging characteristics of a capacitor

1. Objectives:

- To observe charging and discharging characteristics of a capacitor using an oscilloscope
- To verify the time constant in an RC circuit

2. Background:

A capacitor is a passive device that stores energy in it the form of an electric field. It can be charged and discharged through a resistor with the help of a power supply.

During charging, the voltage across the capacitor increases exponentially with time and is given by the following relation

$$V_c(t) = V_0(1 - e^{-\frac{t}{RC}}) \quad (\text{Eqn. 1})$$

Where V_0 is the input voltage and $V_c(t)$ is the voltage across the capacitor at a time t .

Similarly, a charged capacitor can be discharged through a resistor and the voltage across the capacitor at any instant can be found by the following relationship

$$V_c(t) = V_0 e^{-\frac{t}{RC}} \quad (\text{Eqn. 2})$$

Where V_0 is the initial voltage across the capacitor, the term RC in the above equations is called the time constant, τ , measured in terms of seconds.

Time constant is defined as the time required to charge the capacitor, through the resistor, to 63 percent of full charge; or to discharge it to 37 percent of its initial voltage.

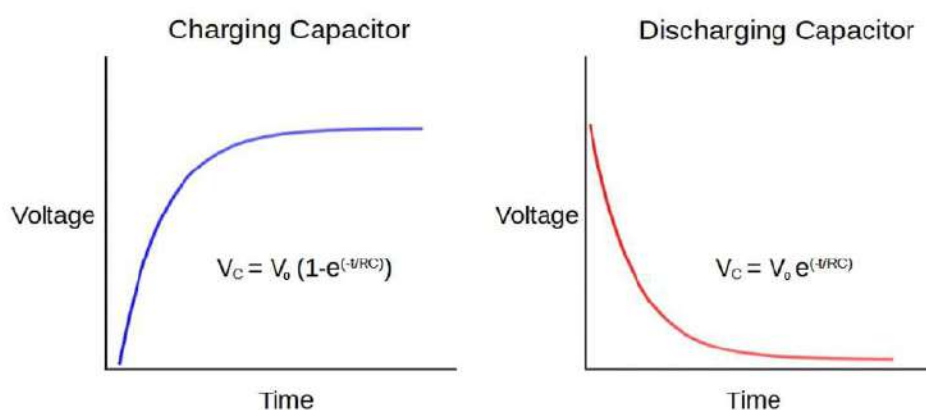


Figure 1: Charging and discharging curve of a capacitor in an RC circuit

The charging phase(left) and the discharging phase(right) of charge, voltage across the capacitor and the current across the resistor is shown below in Figure 2.

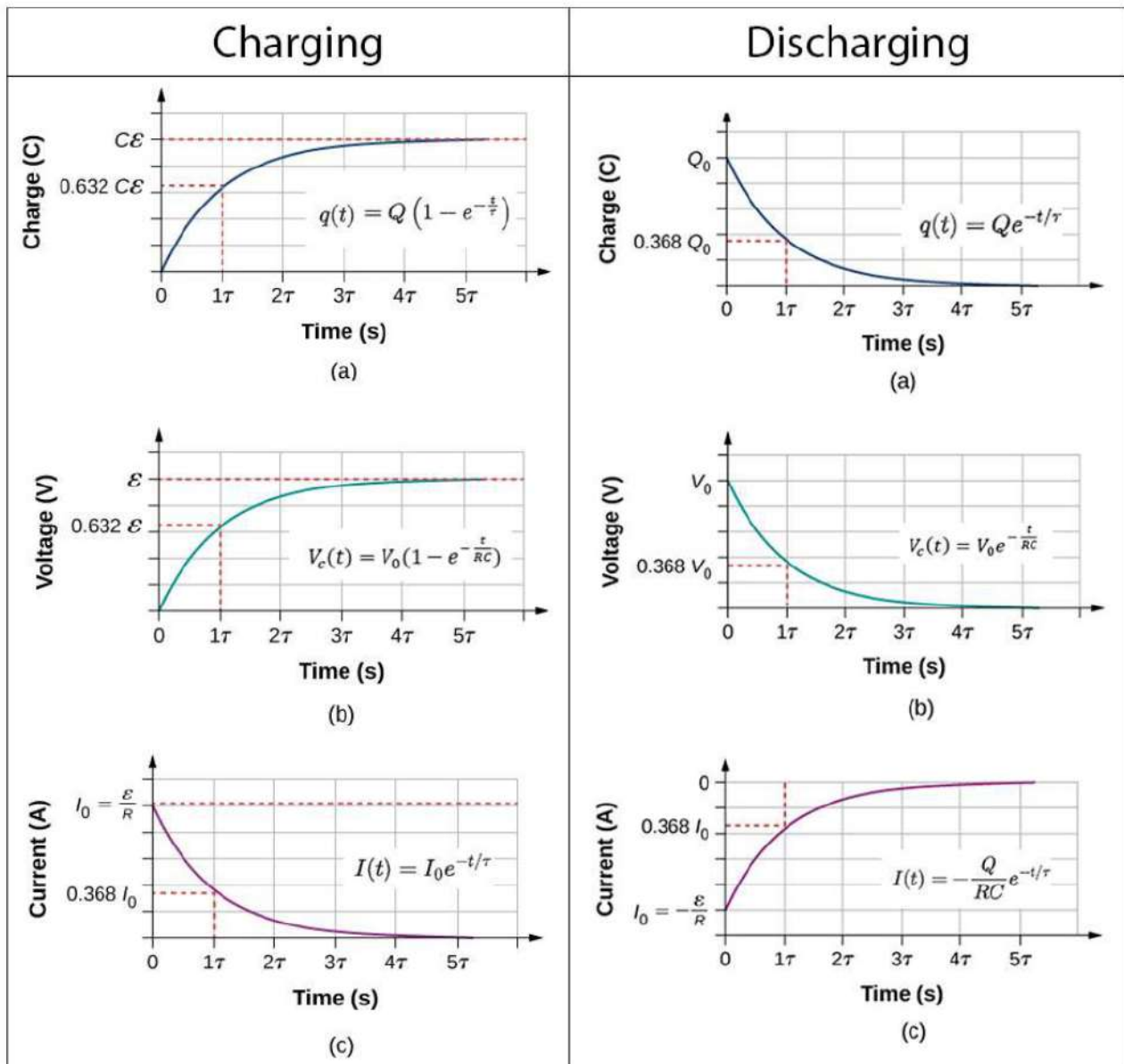


Figure 2. (a) Charge vs Time across Capacitor (b) Voltage vs Time across Capacitor (c) Current vs Time across Resistor

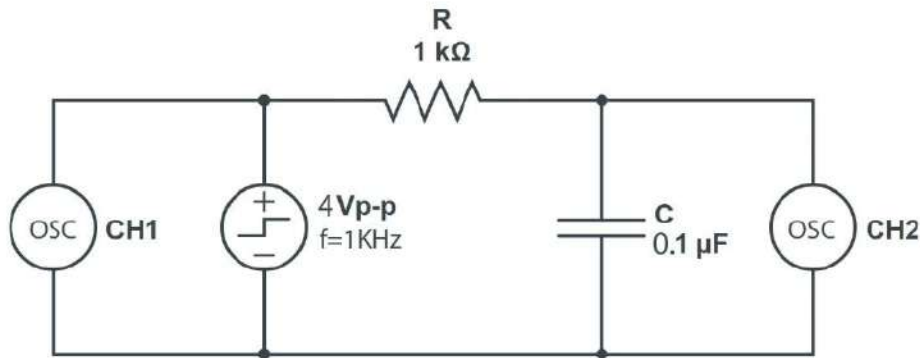


Figure 3: Circuit Diagram for charging and discharging

3. Procedures and Observations:

- Assume that the capacitor is fully discharged. Measure the value of the resistor, $R_{measured} =$
- Connect R and C in series with the AC signal generator. (For reference, please see the circuit given above in figure 2).
- Connect the oscilloscope's channel 1 to the signal generator, choose square wave as an input signal and observe the signal on the oscilloscope.
- Make sure that the channel is set to **DC coupling**.
- Measure the amplitude and period of the input pulse and make sure that the amplitude is set to $4V_{p-p}$ and the time period to 1msec with 50% duty cycle.
- Use the offset knob to raise the signal by 2 V so that the base of the signal is set at 0 V.
- Connect the oscilloscope's channel 2 across the capacitor and observe the output.
- Measure the voltage across the capacitor every 50 μ sec (approximately) and record the data in the following table.
- For calculation of capacitor voltage during discharging, measure the maximum capacitor voltage during charging and use it as V_0 in Eqn. 2.
- Measure the capacitor voltage and record in the following table.

Lab Report:

Date:	
Name of the Students and IDs	(1)
	(2)
	(3)

Data Tables:**Table 1: Time dependent charging characteristic of a capacitor**

Time (μsec)	Calculated V_C (Charging)	Measured V_C	Time (μsec)	Calculated V_C (Charging)	Measured V_C
0					

Table 2: Time dependent discharging characteristic of a capacitor

Time (μsec)	Calculated V_C (Discharging)	Measured V_C	Time (μsec)	Calculated V_C (Discharging)	Measured V_C
0					

Tasks and Questions:

#1: Use data obtained in Table 1 to **plot** capacitor voltage (V_c) vs time (t) graph. From the graph, calculate the time constant and compare it to the theoretical value. Also, label the axes properly.

Time constant (Theoretical): $R_{measured} \times C =$

Time constant (Measured from the graph) =

% of Error =

#2: Use the data obtained in **Table 2** to plot capacitor voltage (V_c) vs time (t) graph. From the above graph, determine the time constant and compare it to the theoretical value. Also, label the axes properly.

Time constant (Theoretical): $R_{measured} \times C =$

Time constant (Measured from the graph) =

% of Error =

#3: Did you notice any difference between theoretical and experimental values of time constants, as obtained from the charging and discharging of capacitors? If any, Explain.

#4: What will happen to the charging curve if the resistance is decreased to 100Ω .

Results:

Discussion: